1. **Watch the video and write notes.**

Watched video and write-down notes.

2. **Create a GitHub account.**

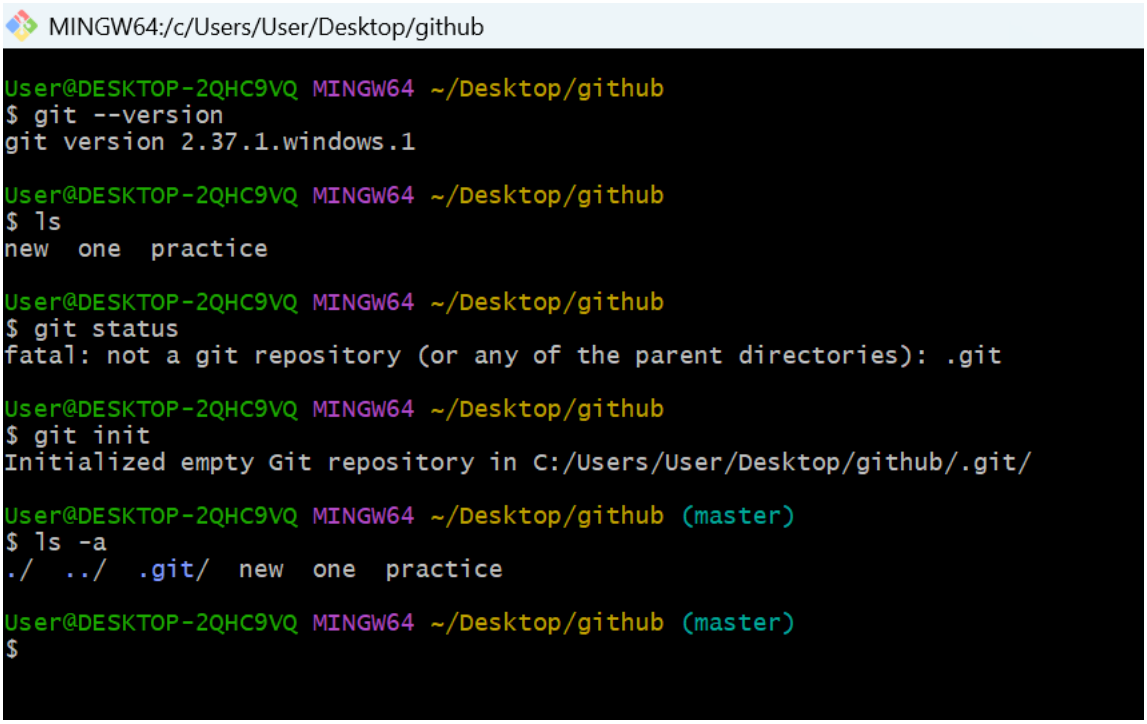
Created GitHub Account.

3. **Set up MFA (Multi-Factor Authentication) for the GitHub account using a mobile app.**

Setup 2 factor authentication via google authenticator.

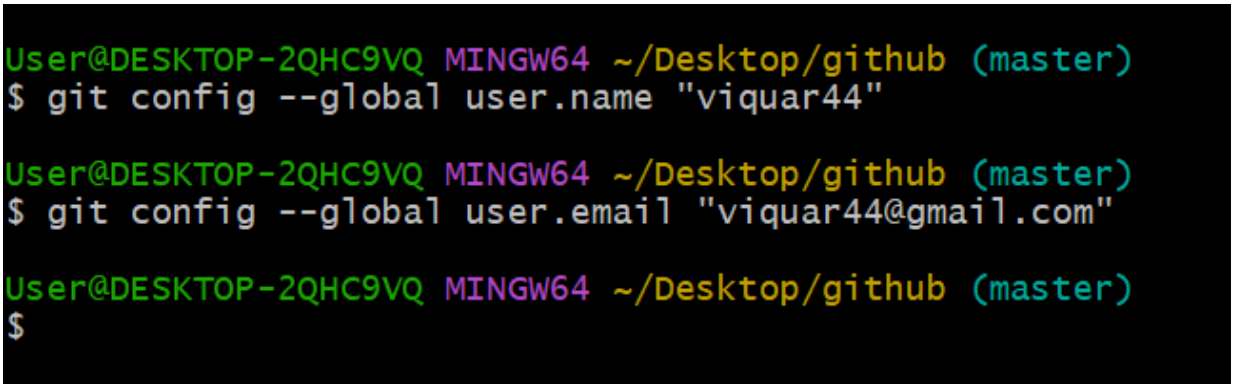
4. **Execute all commands shown in the video.**

Step 1: Open gitbash and check the Git Version (git –version)
Step 2: Create a dir where you want to execute and initiate work and git
Step 3: Initiate git using a command (git init) check the. git (hidden file)



Configure user name and user email

git config - -global user.name “viquar44”
git config - -global user.email “viquar44@gmail.com”



Check the configuration list (git config - -list)

MINGW64:/c/Users/User/Desktop/github

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=viquar44
user.email=viquar44@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$
```

Create a SSH authentication keygen

ssh -keygen

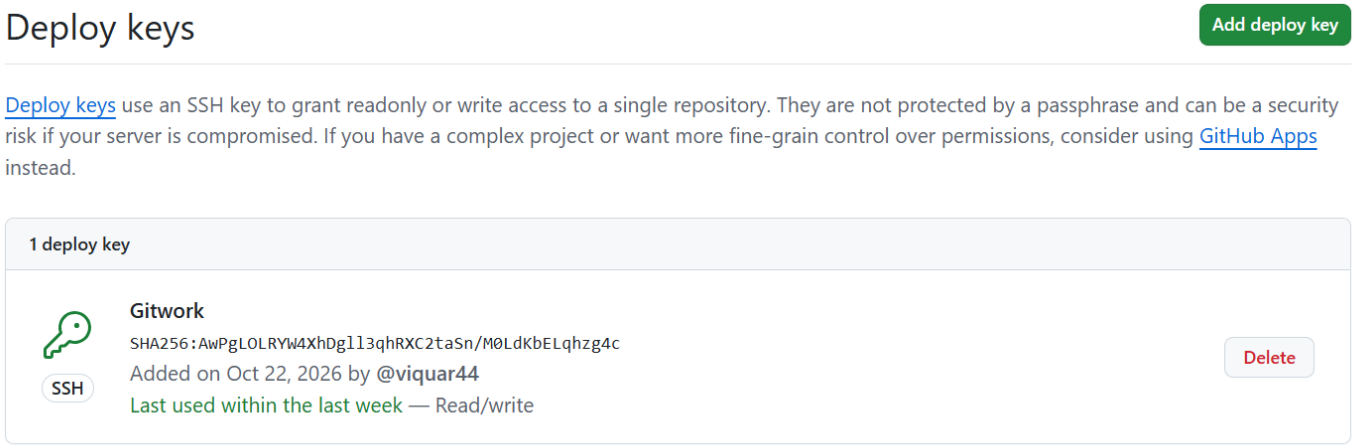
```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/User/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/User/.ssh/id_rsa
Your public key has been saved in /c/Users/User/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:AwPgLOLRYW4XhDgll3qhRXC2taSn/MOLdKbELqhzg4c User@DESKTOP-2QHC9VQ
The key's randomart image is:
+---[RSA 3072]----+
| o*X==           |
| =O=*.o          |
|o.B*.+o          |
|o++.=  o         |
| ..o + oS        |
|  . = *  .       |
| + . = o         |
|E + . . .        |
|.+ .   .         |
+----[SHA256]-----+

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github
$ cat /c/Users/User/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDkZcp8DQXKhLf0gsGivKZgp7x0CWOPKtGH3ivjMTR2I+Oc7VZ5GJLJ6IDfONrkpcMcCraf+KZXkWC8SOY2cI9m0fQ/DvDx3DCBEwQwLor/5htXvdr6NYyrmrGeahc7fk/uLZCN
TtZ+ArRK4BONUwytZIjq9HI9zY5miRNWNKRCHGHDOlHRPIr9SKCWCktFo6xz+qDZrYH16XjDMCvkJntdi7IW7T1oLdCEbzY7jExBiSMqf3Iys1/ZrB2KJUUD/oSOlZSoVnQMJYkMHy54XCLG/fK/9lZUnmr15lOBh1AodSQ2llrR
yLbH727UvplrZ+8HN6uIJyiaOOlijMziq1eOCq9e+VPBN8AWER7VCOSHoCOv5sdyZu7Kr4vNOHv7PFqSksu6B7yXEdlvOlFG/PxFDTb8XHnG2Jou2xCN6gtfXru9T3tkBFiHaua4wl8qjMDzhYyShPuPBwkhL+Wot7xbL13vOllY
VxoX99TmqOPdAfh9IwBvKmKbI9mws1wI4WM= User@DESKTOP-2QHC9VQ

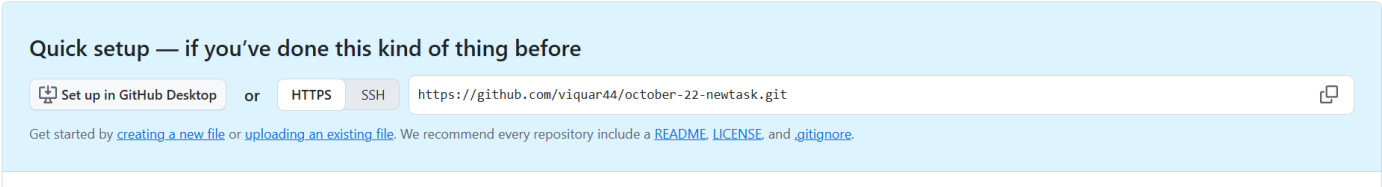
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github
$
```

Copy the keygen and paste to github repo

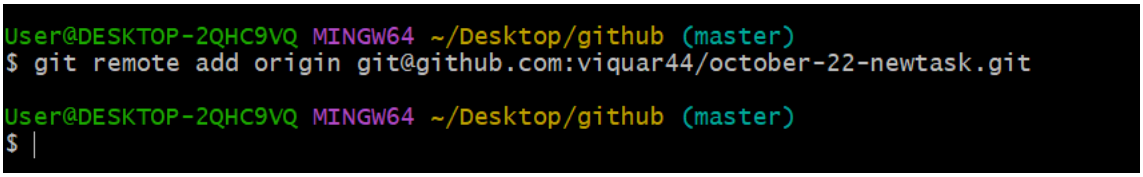
Go to settings of repository → select deploy keys → add deploy key



Connect to repository using code : you will get a link on Github select SSH
git@github.com:viquar44/october-22-newtask.git



Come to git bash and use command (git remote add (paste above link)



If your repository doesn’t have anything you need to apply a command (git branch -M main)

Check the git status and add file to monitor

- ➔ git status
- ➔ git add file name

MINGW64:/c/Users/User/Desktop/github

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        new
        one
        practice

nothing added to commit but untracked files present (use "git add" to track)

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git add new one

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   new
        new file:   one

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        practice


User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git add practice

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git status
On branch master

No commits yet

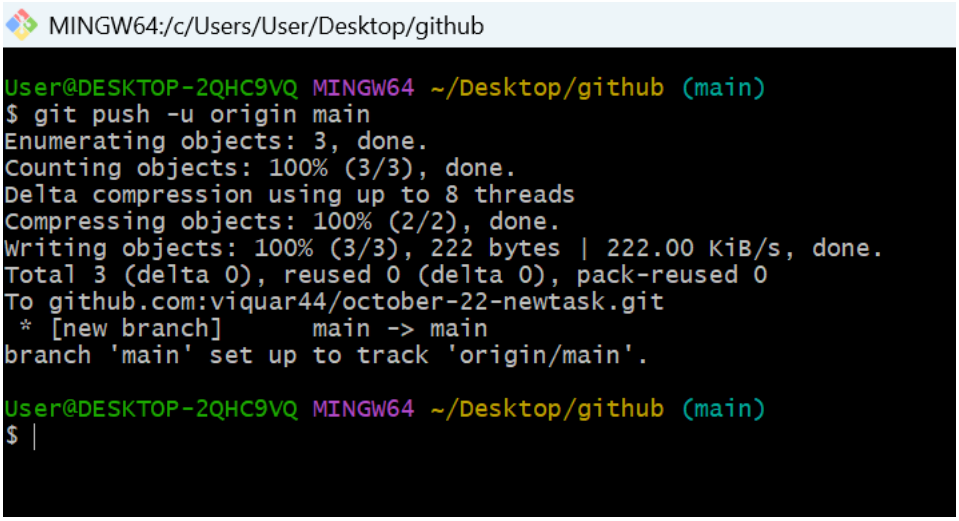
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   new
        new file:   one
        new file:   practice
```

Next step to commit the file using a command **(git commit -m “messege”)**

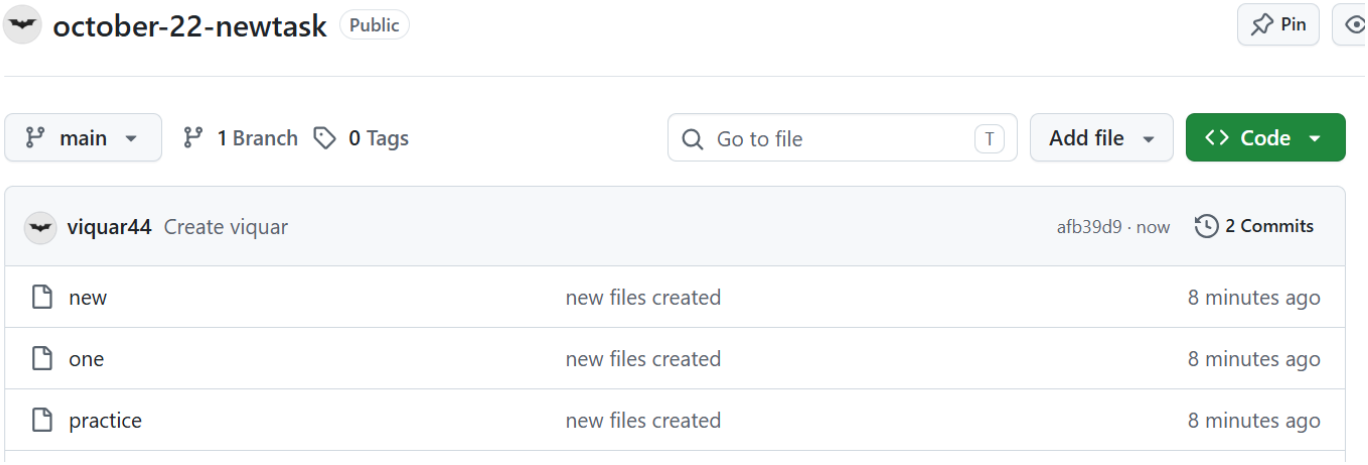
MINGW64:/c/Users/User/Desktop/github

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (master)
$ git commit -m "new files created"
[master (root-commit) da79e34] new files created
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 new
 create mode 100644 one
 create mode 100644 practice
```

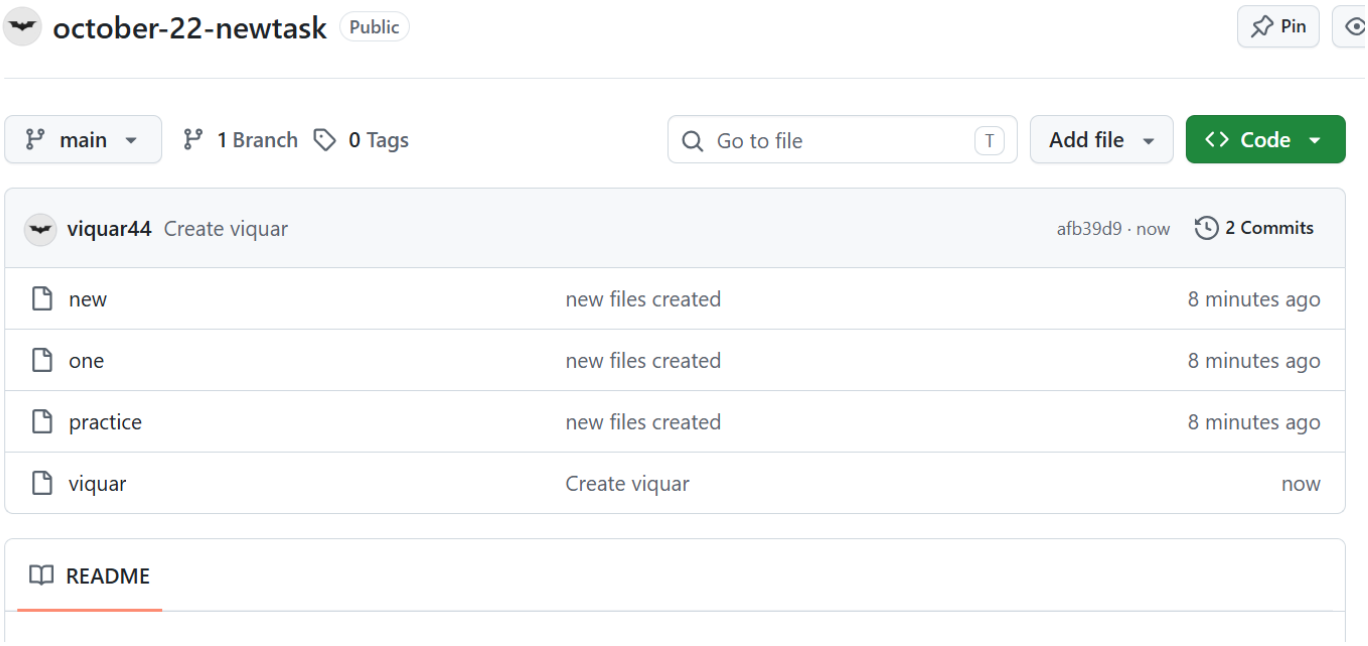
Next step to push to repository using a command **(git push -u origin main)**



Now you can see the file in Gitbash



Then I created a new file in Gitbash name viquar



Now when I try to push it again it rejected because there is new file in Gitbash

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ git push -u origin main
To github.com:viquar44/october-22-newtask.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:viquar44/october-22-newtask.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$
```

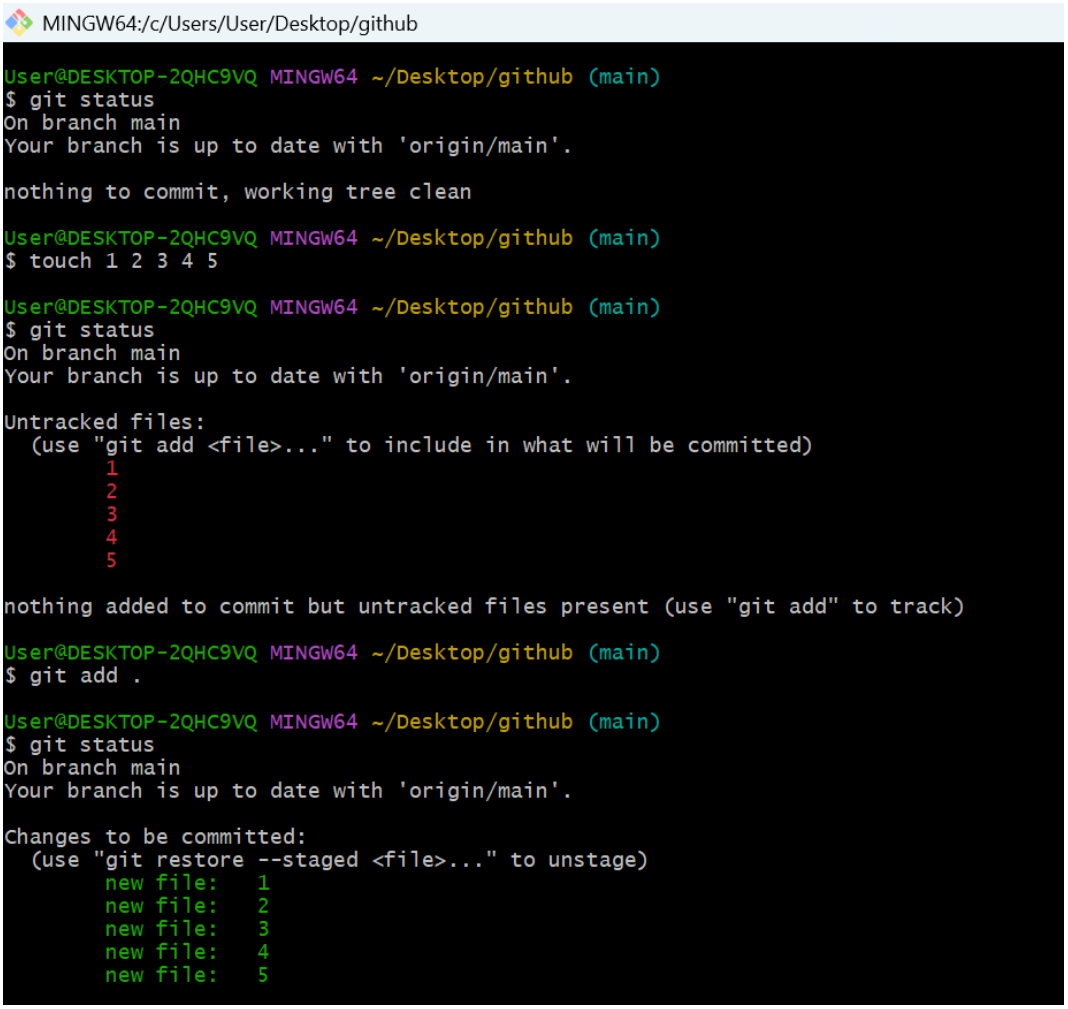
Now need to pull first using a command git pull then check with ls and ll

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 892 bytes | 28.00 KiB/s, done.
From github.com:viquar44/october-22-newtask
   da79e34..afb39d9  main       -> origin/main
Updating da79e34..afb39d9
Fast-forward
 viquar | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 viquar

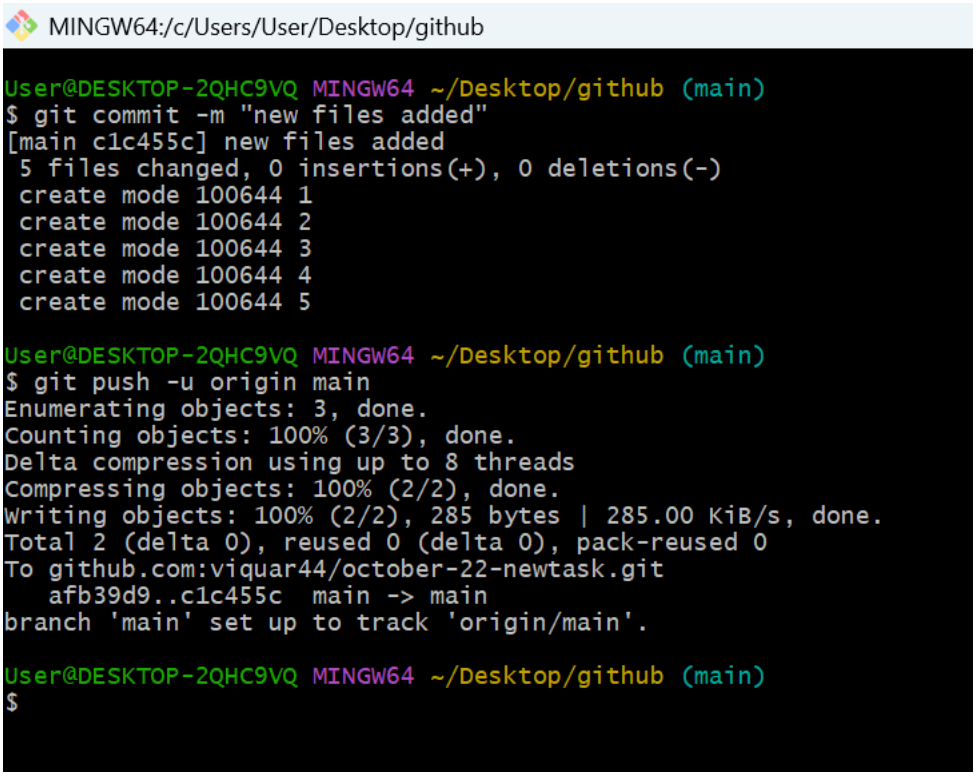
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ ls
new  one  practice  viquar

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ ll
total 1
-rw-r--r-- 1 User 197121 0 Oct 22 15:48 new
-rw-r--r-- 1 User 197121 0 Oct 22 15:48 one
-rw-r--r-- 1 User 197121 0 Oct 22 15:47 practice
-rw-r--r-- 1 User 197121 2 Oct 22 16:21 viquar
```

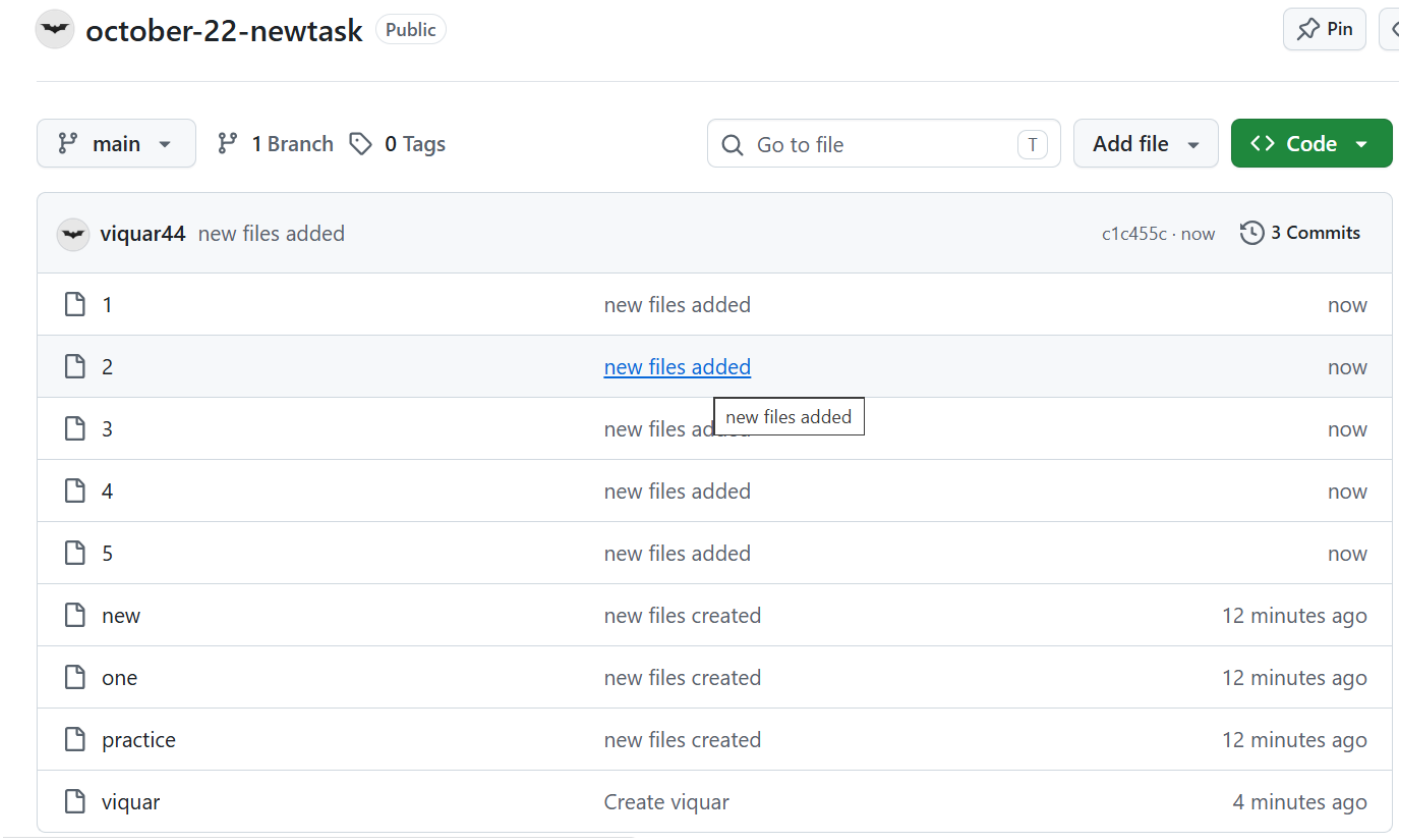
I have created 5 ore file using command touch and check the status and added that 5 files to tracking using a command git add .



The commit files and push (git commit -m “message” and git push -u origin main)



Now you can check that files in Github



I have created new file and add to track but doesn't want to commit use command (git reset)

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ touch ten

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ ls
1  2  3  4  5  new  one  practice  ten  viquar

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ten

nothing added to commit but untracked files present (use "git add" to track)

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ git add ten
```

```
User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ git reset

User@DESKTOP-2QHC9VQ MINGW64 ~/Desktop/github (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ten

nothing added to commit but untracked files present (use "git add" to track)
```